

Methods for calculating return

money weighted (MWR)

$$V_0(1+i)^T, \sum C_t(1+i)^{T-t} = V_T$$

- absolute measure

problems:

- not good for comparing fund managers
 - ↳ RR influenced by CF & time
- payments in & out not in control of fund manager

Reasonable:

- no large CF
- RR remains stable

Time-weighted (TWR)

$$(1+i)^T = \frac{V_{01}}{V_0} \times \frac{V_{12}}{V_{01} + CF_1} \times \dots \times \frac{V_T}{V_{T-1} + CF_{T-1}}$$

- overcomes MWR problem

- compare fund managers
- timing & size does not distort

calculate:

- assess fund value when CF in or out
- calculate return achieved for each period between CF
- link to give TWR

impracticality:

- requires large volume of data

→ excluding investment profits

→ before CF

MWR + TWR: similar:

- CF are small relative to the fund
- RR is stable overtime

Linked Internal Rate of Return (LIAR)

Approximation of TWR

↳ quarterly sub-intervals

PROCESS:

- fund value at various dates
- calculate MWR for each inter-valuation period
- link MWR to get LIAR

when is it a good ~:

- CF small relative to the size of the fund
- RR stable over intervaluation period

Compare portfolio performance with index:

① Compare actual value of portfolio at end of defined period with value that would have been achieved had the initial value + subsequent funds been invested in the same way as the index

② Compare TWR (or LIAR)

Performance Attribution



↪ Sector

F_{AN} = fund value from actual sector + stocks

F_{NN} = fund value from notional sector + stocks

Overall PIL = $F_{AN} - F_{NN}$

Stock selection profit = $F_{AN} - F_{NN}$

Sector selection profit = $F_{AN} - F_{NN}$

T (return) → No weights

Contribution to stock selection profit = $A_{\text{sector weight}} \times (A_{\text{sector return}} - N_{\text{sector return}})$

Contribution to sector selection profit = $(A_{\text{sector weight}} - N_{\text{sector weight}}) \times (N_{\text{sector return}} - \text{Benchmark return})$

CHAPTER 16: PERFORMANCE MEASUREMENT

$$R_{1A} = \frac{MV_1 - MV_0}{MV_0}$$

→ Adjust for CF

CF at start

$$r_t = \frac{MV_1 - (MV_0 + CF)}{MV_0 + CF}$$

CF at end

$$r_t = \frac{(MV_1 - CF) - MV_0}{MV_0}$$

Comparing portfolio performance with benchmark:

Assumptions:

- how new money + investment income are to be invested
- how often benchmark is rebalanced

complications:

- contributions
- taxes
- investment income
- expenses

Risk-adjusted Performance Measures

Measures

Allows: Risk → variance

Ignore: < actual risk → fail to meet objective / payments
downside risk → fail to achieve specified target return

Limitations

- assumes CAPM = ✓
- no actuarial / downside risk
- time → not repeat in future
- grow/ret fee/tax?

Standard Deviation

- All of investor's assets
- ⊕ Capital market line
→ hold well-diversified portf.
of risky assets = efficient

σ → measures how well-diversified whole portfolio is

→ how good manager is at picking stocks producing excess return relative to their p

Beta

- Part of investor's assets
- ⊕ Security market line
→ all portfolios, efficient or otherwise

β → measures risk wrt well diversified portfolio

→ adjusting return using β = explains how good manager is at picking out-performing stocks given systematic risk

Treynor measure

$$T = \frac{R_p - r}{\beta_p}$$

Malme measure

$$J = \frac{R_p - r}{\sigma_p}$$

Jensen measure

$$R_b = r + \beta_p(R_m - r)$$

$$J = R_p - R_b$$

Pre-specified std deviation

$$R_b = r + \frac{R_m - r}{\sigma_m} \sigma_p$$

$$DM = R_p - R_b$$

Outperformance as proportion predicted by CAPM

Outperformance as absolute amount exceeding that predicted by CAPM

Limitations

Projection of past results

- caveat → past not rep future

- past events may not recur in the future

- random element in investment returns that can be difficult to determine in performance measure

Risk adjusted Risks

- difficulty in allowing for risks
Past risks ≠ future

Time scale

- internal manag to avoid short-termism

- frequently not high enough to establish + correct problems

- too short measurement periods
→ suspicious accuracy

Impact fund manager's behaviour

- incentives

- knowledge on when + how to be assessed influences investment strategies

- strategies ≠ fund's best interests

Costs

cost vs ben

- many measures
- avoid subjective

- valuations = difficult, subjective, time consuming

- detailed + frequent calculations may be based on subjective val
→ inappropriate

Differing fund objectives

- find with same
- adjust for diff

- invalid comparison if different objectives + constraints

- different tax position

Use of performance measures (PRIK)

- improve future performance
- compare rate achieved vs target
- compare performance to another portfolio, index or benchmark
- appraise + remunerate investment managers

Relative performance

Index

Ad: easy

↳ data < readily available
accurate

Disad: index may not be appropriate

Benchmark

Ad: avoids giving managers conflicting objectives

↳ reflects fund's liabilities

Other portfolios

Appropriate if funds have the same < objectives
factors influencing investment strategy

Ad: indicates cost + benefit
Disad: inappropriate if diff obj.